

Challenge-2

Objectives:

- Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.
- Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

Note:

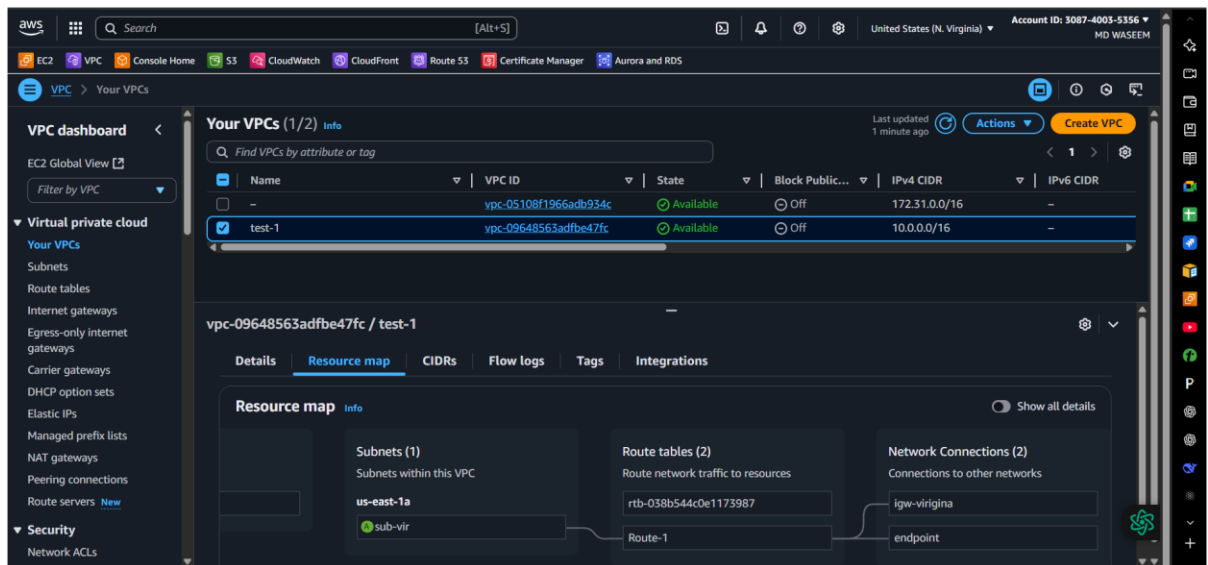
- Create 4 VPCs in 4 different regions and set up Transit Gateway.
- Do not use default VPCs.
- Prepare a document with all the screenshots and push it to a GitHub repo, then share the URL.

- **Step1: 4 Different IP Address**
- 1.) VPC-A (N. Virginia, us-east-1) → 10.0.0.0/16
- 2.) VPC-B (OHIO -us-east-2) → 10.1.0.0/16
- 3.) VPC-C (California, us-west-1) → 10.2.0.0/16
- 4.) VPC-D (Oregon, us-west-2) → 10.3.0.0/16

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- **Step2: Create 4 VPCS in Different regions**
- Go to VPC Console → Create VPC → VPC Settings → VPC Only → Name
- VPC-1 N.Virginia → IPv4 CIDR → 10.0.0.0/16 → Create VPC
- Subnets:
- Create 4 Private subnets in different regions
- Route Tables:
- Create a 4 Route Table for each set of subnets and associate them
- Attached IGW to One VPC (N.Virginia)

Vpc of virginia



Challenge-2

Vpc of ohio

The screenshot shows the AWS VPC console for the 'United States (Ohio)' region. The 'Your VPCs' table lists two VPCs: 'test2' (vpc-0da7f479ef828667e) and another VPC (vpc-0c4e6fee294d998e1). The 'test2' VPC is selected, and the 'Resource map' tab is active. The resource map shows the VPC's internal structure, including subnets, route tables, and network connections.

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
-	vpc-0c4e6fee294d998e1	Available	Off	172.31.0.0/16	-
test2	vpc-0da7f479ef828667e	Available	Off	10.1.0.0/16	-

Resource map for vpc-0da7f479ef828667e / test2:

- Subnets (1):** Subnets within this VPC. Includes 'us-east-2a' and 'vpc-ohio-pri'.
- Route tables (2):** Route network traffic to resources. Includes 'rtb-0d3dc2b82c1a6c409' and 'Route-ohio'.
- Network Connections (1):** Connections to other networks. Includes 'end-2'.

Vpc of california

The screenshot shows the AWS VPC console for the 'United States (N. California)' region. The 'Your VPCs' table lists two VPCs: 'test-3' (vpc-017533921c581bbd6) and another VPC (vpc-0ab98194354bc3910). The 'test-3' VPC is selected, and the 'Resource map' tab is active. The resource map shows the VPC's internal structure, including subnets, route tables, and network connections.

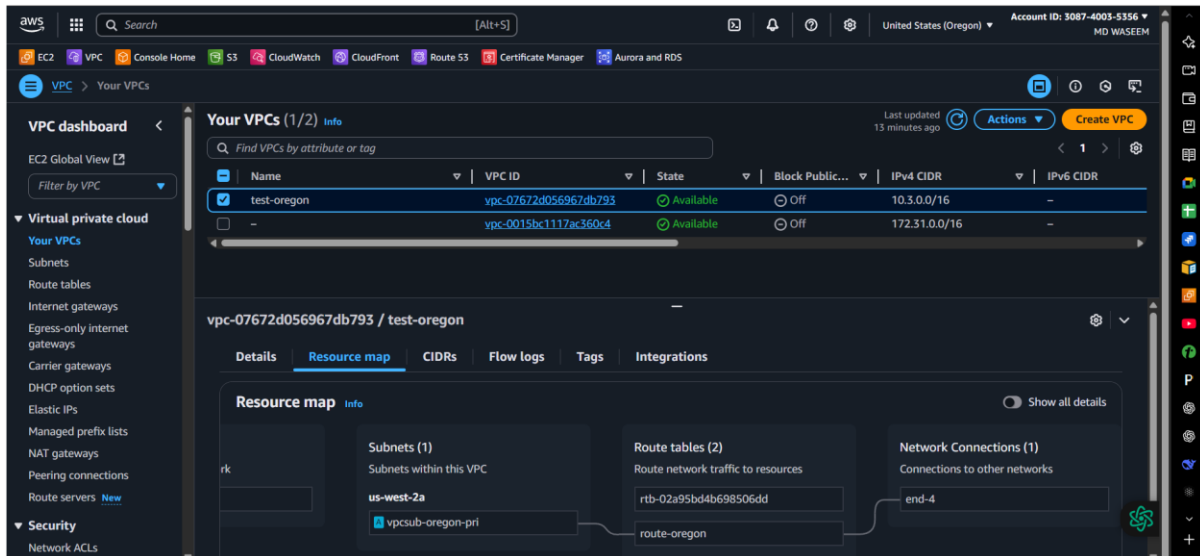
Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
test-3	vpc-017533921c581bbd6	Available	Off	10.2.0.0/16	-
-	vpc-0ab98194354bc3910	Available	Off	172.31.0.0/16	-

Resource map for vpc-017533921c581bbd6 / test-3:

- Subnets (1):** Subnets within this VPC. Includes 'us-west-1a' and 'vpcsub-california-pri'.
- Route tables (1):** Route network traffic to resources. Includes 'rtb-0329caGea8284d68b'.
- Network Connections (1):** Connections to other networks. Includes 'end-3'.

Challenge-2

Vpc of oregon



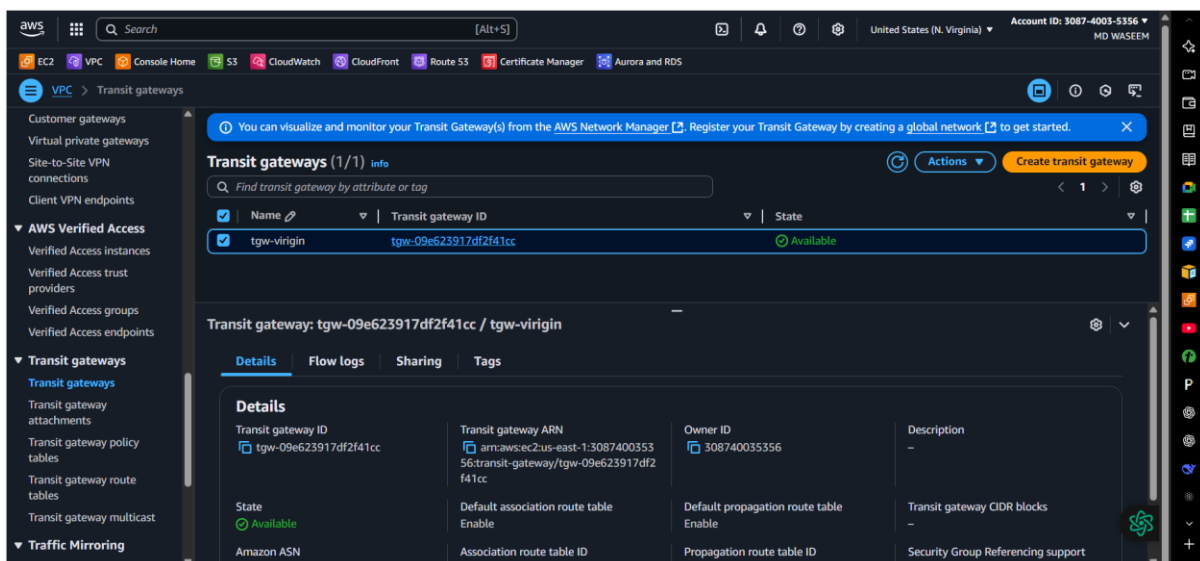
Here we are creating vpc subnet route and its internet gateway(only in primary region)

Step 3: Create Transit Gateways:

a. Create a Transit Gateway

VPC Console → Transit Gateways → Create transit gateway

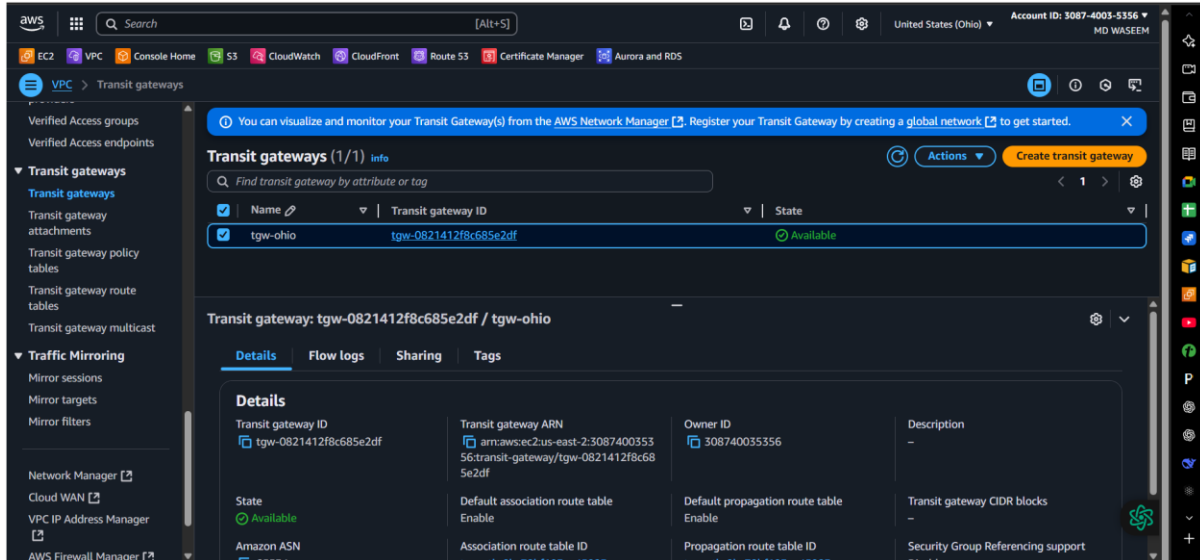
→ Transit of virginia



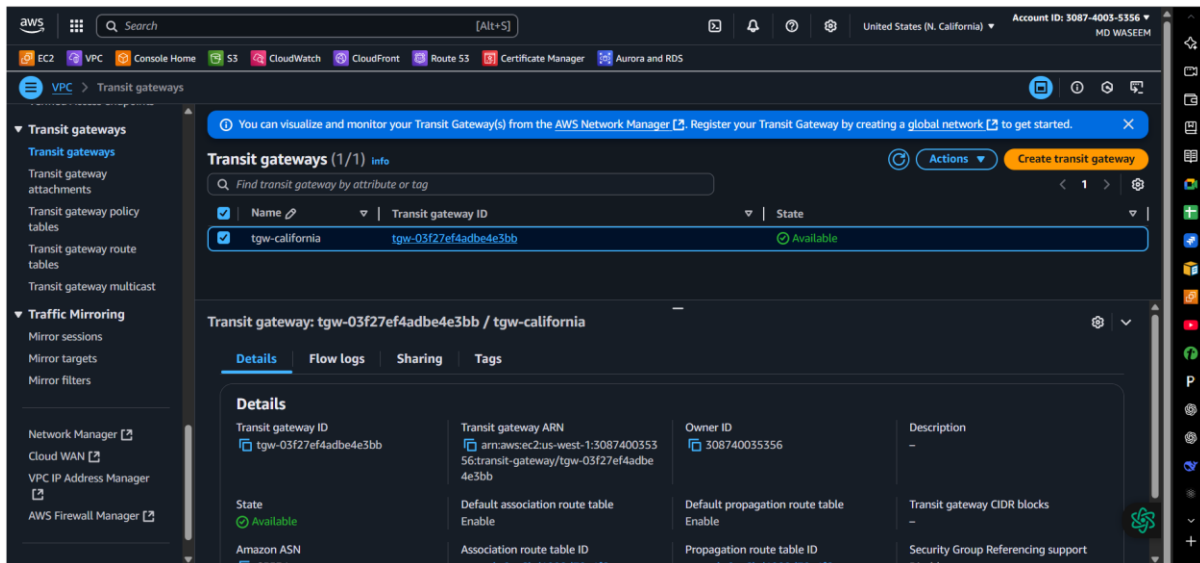
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Now we have to create transit gateway in all region and also need to add transit gateway attachment and finally need to add peering by adding transit gateway id of all region at a time

➔ Transit gateway of ohio

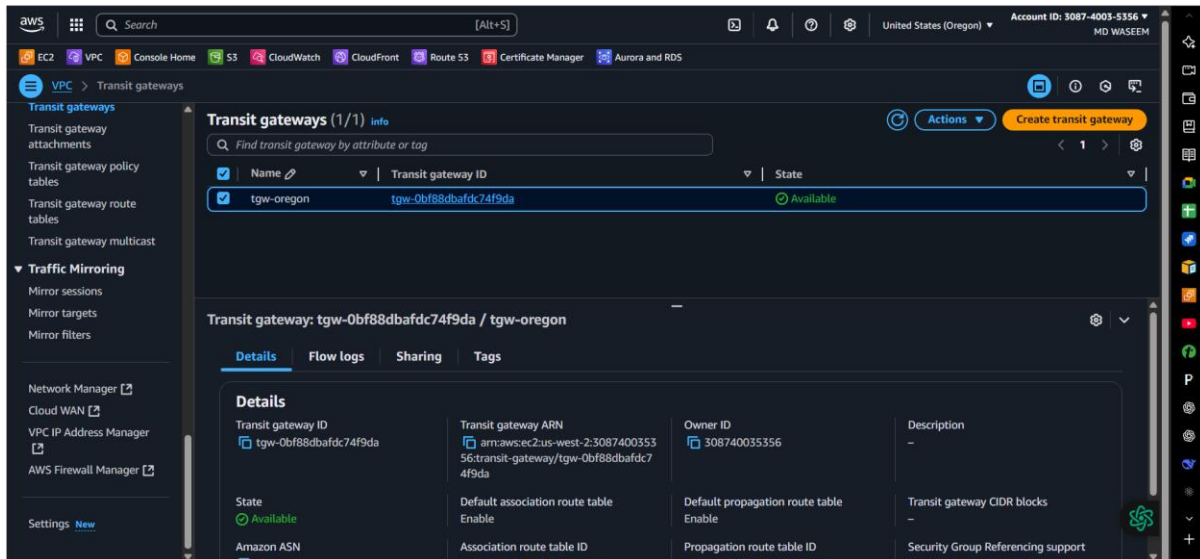


➔ Transit gateway from California



Challenge-2

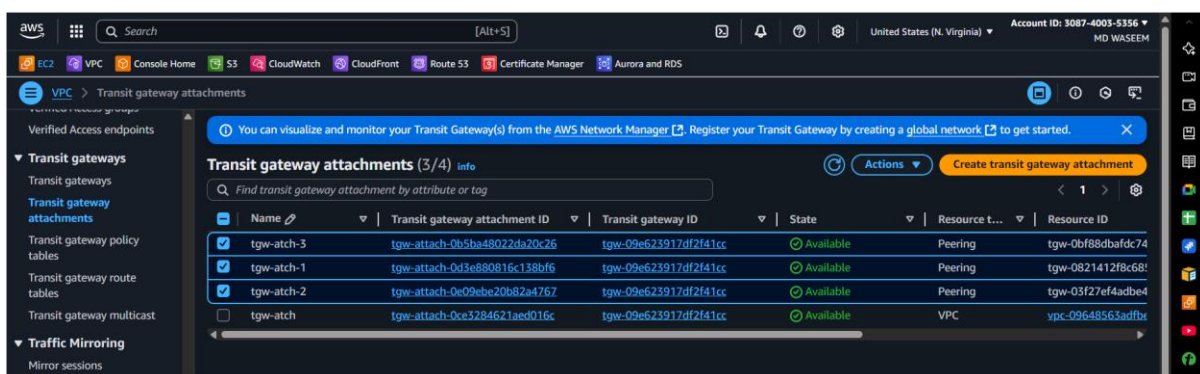
→ Transit gateway from Oregon



→ Sending request from primary region (Virginia)

→ The one which selected are peering connected with another region.

- Go to Transit Gateway Attachments → Create transit gateway attachment
- Attachment type: VPC
- Choose VPC: Select the regional VPC
- Subnets: Select private subnets
- Create attachment



Challenge-2

→ Peering accepted from ohio

The screenshot shows the AWS Management Console for the 'United States (Ohio)' region. The left sidebar is expanded to 'VPC' > 'Transit gateway attachments'. The main panel displays a table of 'Transit gateway attachments (1/2)'. The table has columns: Name, Transit gateway attachment ID, Transit gateway ID, State, Resource type, and Resource ID. Two attachments are listed: 'tgw-attach-0d3e880816c138bf6' (Peering) and 'tgw-attach-08e981cd4c44c282f' (VPC). The VPC attachment is selected. Below the table, the 'Details' tab for 'tgw-attach-08e981cd4c44c282f / tgw-attach' is shown. It includes fields for Transit gateway attachment ID, Transit gateway ID, Transit gateway owner ID, State (Available), Resource owner ID, Resource type (VPC), Resource ID, DNS support (Enable), Security Group Referencing support (Enable), Appliance Mode support (Disable), Subnet IDs (subnet-0c8018b7af1dde55a), Association state (Associated), and Association route table ID.

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
tgw-attach-0d3e880816c138bf6	tgw-attach-0d3e880816c138bf6	tgw-0821412f8c685e2df	Available	Peering	tgw-09e623917d
tgw-attach	tgw-attach-08e981cd4c44c282f	tgw-0821412f8c685e2df	Available	VPC	vpc-0da7f479ef8

Transit gateway attachment: tgw-attach-08e981cd4c44c282f / tgw-attach

Details

Transit gateway attachment ID	Transit gateway ID	Transit gateway owner ID	Appliance Mode support
tgw-attach-08e981cd4c44c282f	tgw-0821412f8c685e2df	308740035356	Disable
State	Resource owner ID	DNS support	Subnet IDs
Available	308740035356	Enable	subnet-0c8018b7af1dde55a
Resource type	Resource ID	Security Group Referencing support	Association state
VPC	vpc-0da7f479ef828667e	Enable	Associated
Association route table ID	IPv6 support		

→ peering accepted from California

The screenshot shows the AWS Management Console for the 'United States (N. California)' region. The left sidebar is expanded to 'VPC' > 'Transit gateway attachments'. The main panel displays a table of 'Transit gateway attachments (1/2)'. The table has columns: Name, Transit gateway attachment ID, Transit gateway ID, State, Resource type, and Resource ID. Two attachments are listed: 'tgw-attach-0e09e8e20b82a4767' (Peering) and 'tgw-attach-0978030b42f50848c' (VPC). The VPC attachment is selected. Below the table, the 'Details' tab for 'tgw-attach-0978030b42f50848c / tgw-attach' is shown. It includes fields for Transit gateway attachment ID, Transit gateway ID, Transit gateway owner ID, State (Available), Resource owner ID, Resource type (VPC), Resource ID, DNS support (Enable), Security Group Referencing support (Enable), Appliance Mode support (Disable), Subnet IDs (subnet-0fe6276468056c207), Association state (Associated), and Association route table ID.

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
tgw-attach-0e09e8e20b82a4767	tgw-attach-0e09e8e20b82a4767	tgw-03f27ef4adbede3bb	Available	Peering	tgw-09e623917d
tgw-attach	tgw-attach-0978030b42f50848c	tgw-03f27ef4adbede3bb	Available	VPC	vpc-017533921c

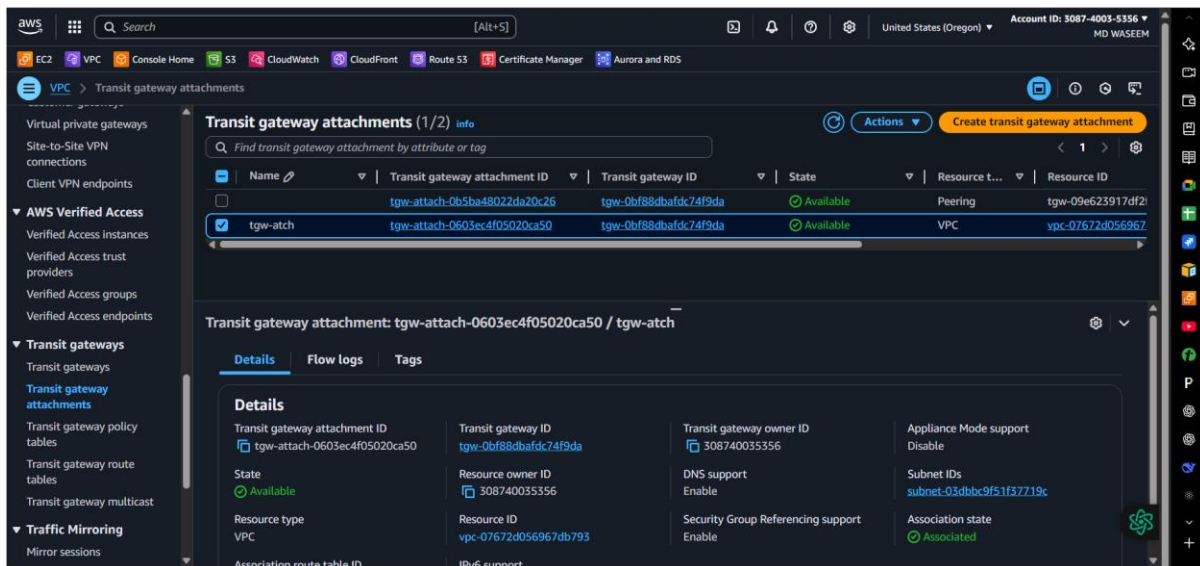
Transit gateway attachment: tgw-attach-0978030b42f50848c / tgw-attach

Details

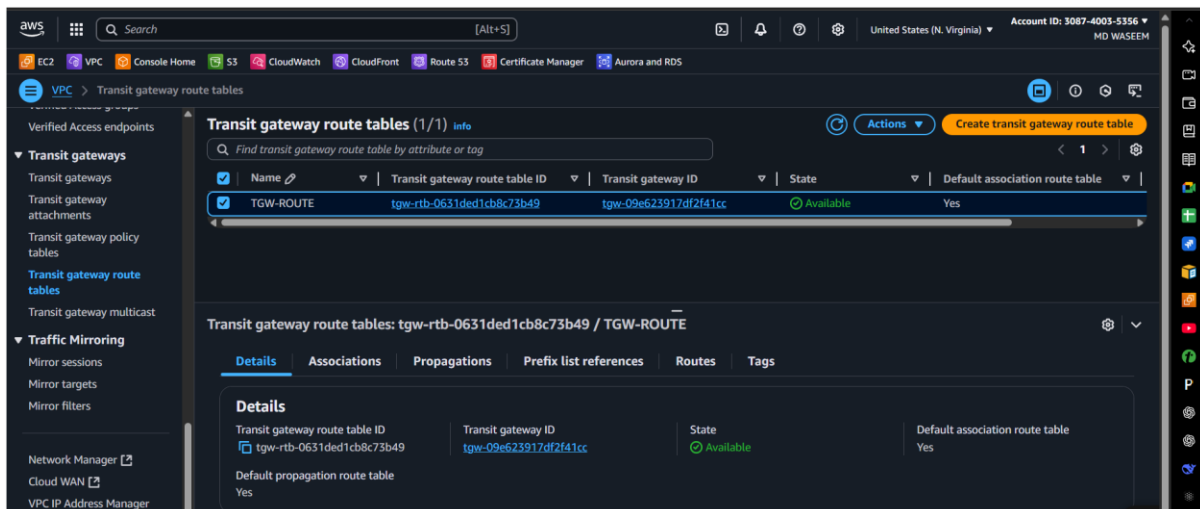
Transit gateway attachment ID	Transit gateway ID	Transit gateway owner ID	Appliance Mode support
tgw-attach-0978030b42f50848c	tgw-03f27ef4adbede3bb	308740035356	Disable
State	Resource owner ID	DNS support	Subnet IDs
Available	308740035356	Enable	subnet-0fe6276468056c207
Resource type	Resource ID	Security Group Referencing support	Association state
VPC	vpc-017533921c581bbd6	Enable	Associated
Association route table ID	IPv6 support		

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→ Here we have accepted from Oregon



Here in transit gateway route table →Route tab →VPC CIDR range of all region except the current region (Virginia)



Challenge-2

→ Here is the added CIRD ranges of all other region. (virginia)

. Associate VPC Attachment with TGW Route Table

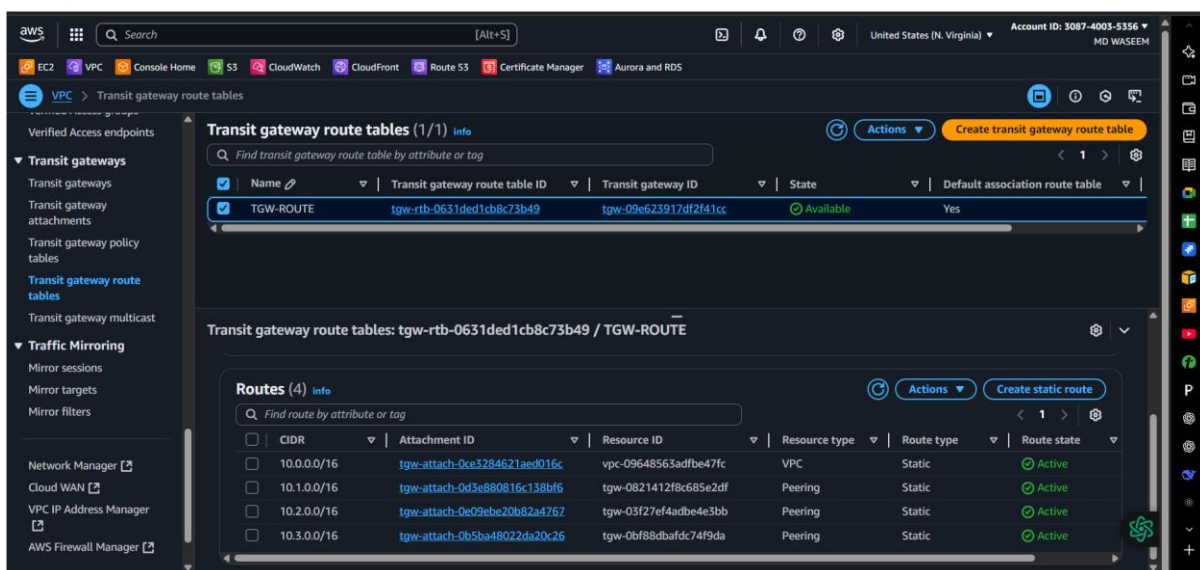
Navigate: Transit Gateway Route Tables

Select TGW's route table

Choose Associations → Create association

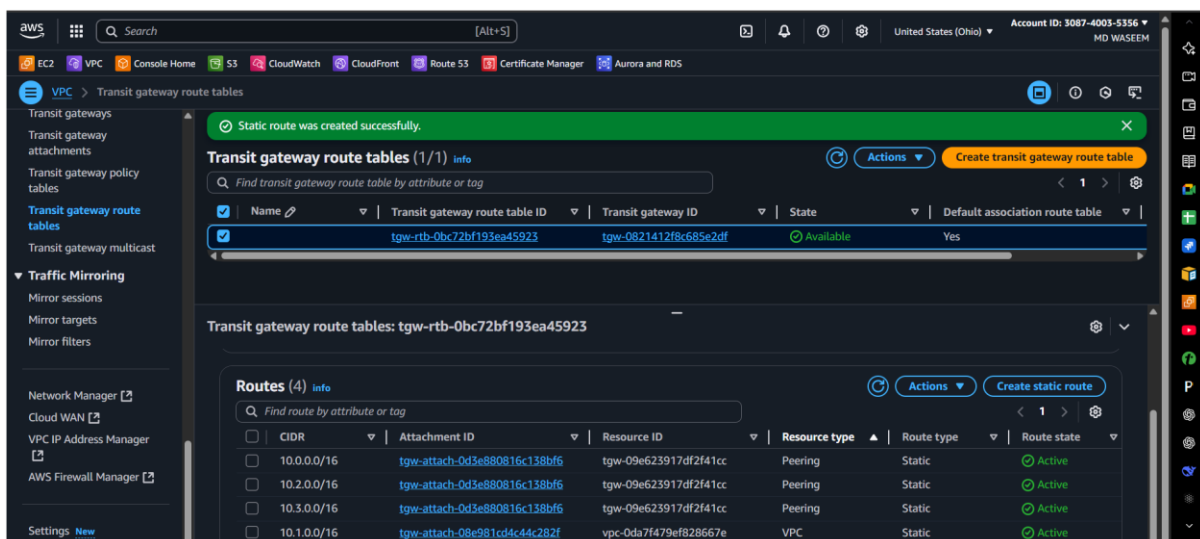
Pick the VPC attachment.

d. Update VPC Route Tables



→ here we add the CIRD of all other region except the current region

Route of ohio



Challenge-2

→Route of California here we have add of CIDR range of other except the current

The screenshot shows the AWS Management Console for the 'United States (N. California)' region. The left sidebar is expanded to 'VPC' > 'Transit gateway route tables'. The main content area displays 'Transit gateway route tables (1/1)' with a table containing one entry: 'tgw-rtb-0aa6bd1902d70a4f6' associated with 'tgw-03f27ef4adbe4e3bb' and in an 'Available' state. Below this, the 'Routes (4)' for this table are listed:

CIDR	Attachment ID	Resource ID	Resource type	Route type	Route state
10.0.0.0/16	tgw-attach-0e09ebe20b82a4767	tgw-09e623917df2f41cc	Peering	Static	Active
10.1.0.0/16	tgw-attach-0e09ebe20b82a4767	tgw-09e623917df2f41cc	Peering	Static	Active
10.2.0.0/16	tgw-attach-0978030b42f50848c	vpc-017533921c581bbd6	VPC	Static	Active
10.3.0.0/16	tgw-attach-0e09ebe20b82a4767	tgw-09e623917df2f41cc	Peering	Static	Active

→Route from Oregon add CIRD of all region except the current region.

The screenshot shows the AWS Management Console for the 'United States (Oregon)' region. The left sidebar is expanded to 'VPC' > 'Transit gateway route tables'. The main content area displays 'Transit gateway route tables (1/1)' with a table containing one entry: 'tgw-rtb-067cca30391e45833' associated with 'tgw-0bf88dbafdc74f9da' and in an 'Available' state. Below this, the 'Routes (4)' for this table are listed:

CIDR	Attachment ID	Resource ID	Resource type	Route type	Route state
10.0.0.0/16	tgw-attach-0b5ba48022da20c26	tgw-09e623917df2f41cc	Peering	Static	Active
10.1.0.0/16	tgw-attach-0b5ba48022da20c26	tgw-09e623917df2f41cc	Peering	Static	Active
10.2.0.0/16	tgw-attach-0b5ba48022da20c26	tgw-09e623917df2f41cc	Peering	Static	Active
10.3.0.0/16	tgw-attach-0603ec4f05020ca50	vpc-07672d056967db793	VPC	Static	Active

Challenge-2

→ Now vpc route add the same CIRD of other region add in (Destination) and in transit gateway (target)

→ Do the same in all other region by add CIDR range and select transit gateway In target.

The screenshot shows the AWS Management Console interface for the 'Route tables' section. The 'Route-1' table is selected, and its routes are displayed. The routes include a transit gateway target for the 10.3.0.0/16 destination.

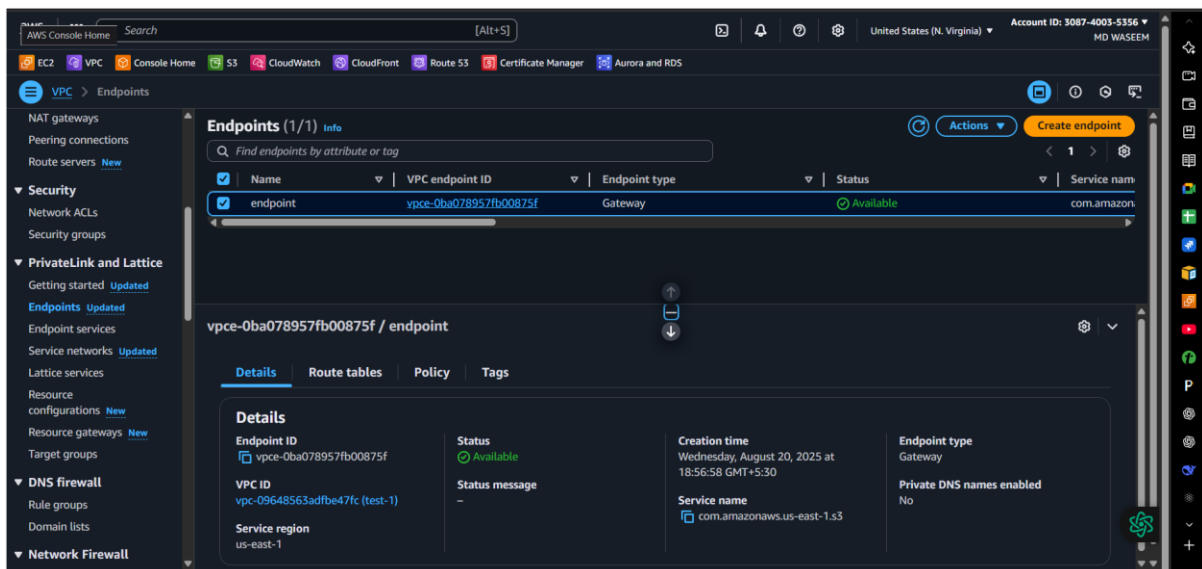
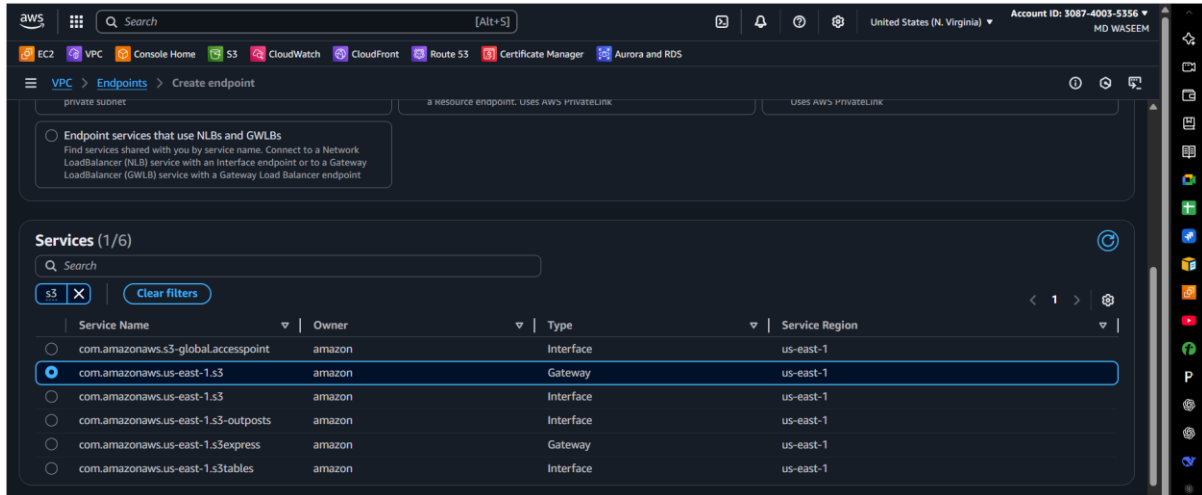
Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-0f50e0b55c1526db0	-	-	Yes	vpc-05108f1966adb...
-	rtb-038b544c0e1173987	-	-	Yes	vpc-09648563adfb...
Route-1	rtb-07f2e8e5e562e72e	subnet-073f958cb49148...	-	No	vpc-09648563adfb...

Destination	Target	Status	Propagated	Route Origin
pl-63a5400a	vpce-0ba078957fb00875f	Active	No	Create Route
0.0.0.0/0	lgw-03cec739e7b3e5c5d	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
10.1.0.0/16	tgw-09e623917df2f41cc	Active	No	Create Route
10.2.0.0/16	tgw-09e623917df2f41cc	Active	No	Create Route
10.3.0.0/16	tgw-09e623917df2f41cc	Active	No	Create Route

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Now we need to create Endpoint all of region.

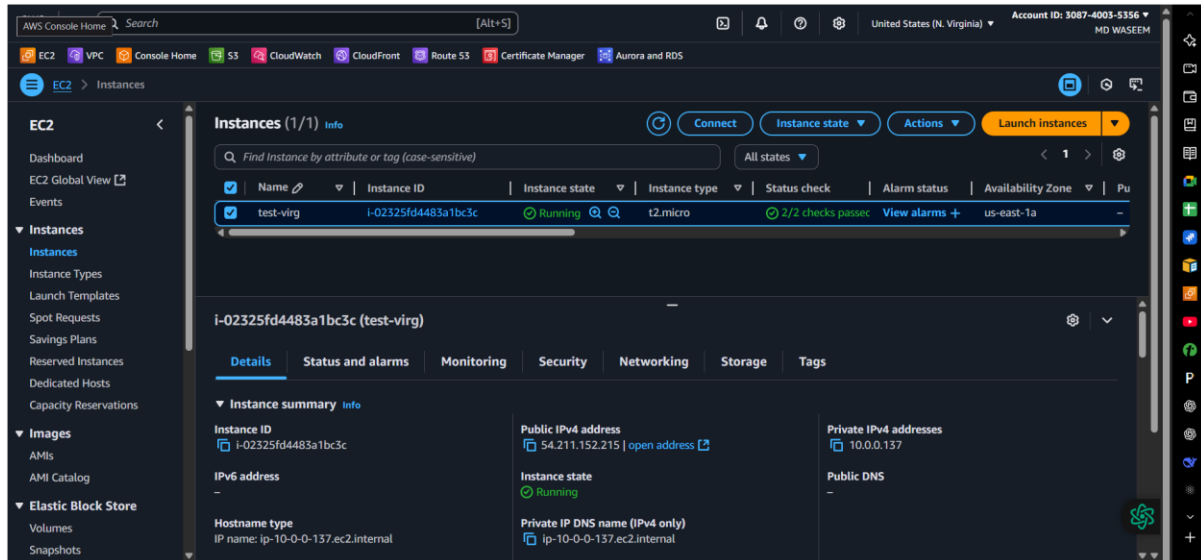
Create Endpoint add this in →service→search of s3 (type) gateway.



Challenge-2

Now create instances for all 4 region

Instances of virginia

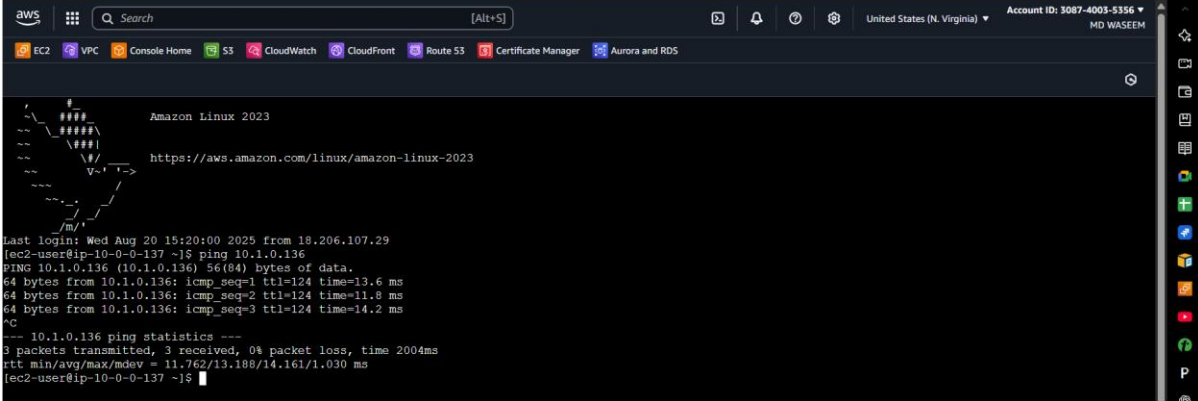


Now create all other 3 region instances same as the above

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Now here we all set to make ping (check connection)

Connecting from Virginia to ohio 



```
aws
Search [Alt+S]
United States (N. Virginia) Account ID: 3087-4003-5356 MD WASEEM
EC2 VPC Console Home S3 CloudWatch CloudFront Route 53 Certificate Manager Aurora and RDS

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Wed Aug 20 15:20:00 2025 from 18.206.107.29
[ec2-user@ip-10-0-0-137 ~]$ ping 10.1.0.136
PING 10.1.0.136 (10.1.0.136) 56(84) bytes of data:
64 bytes from 10.1.0.136: icmp_seq=1 ttl=124 time=13.6 ms
64 bytes from 10.1.0.136: icmp_seq=2 ttl=124 time=11.8 ms
64 bytes from 10.1.0.136: icmp_seq=3 ttl=124 time=14.2 ms
^C
--- 10.1.0.136 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 11.762/13.188/14.161/1.030 ms
[ec2-user@ip-10-0-0-137 ~]$
```

Connecting from Virginia to California 

```
[ec2-user@ip-10-0-0-137 ~]$ ping 10.2.0.234
PING 10.2.0.234 (10.2.0.234) 56(84) bytes of data.
64 bytes from 10.2.0.234: icmp_seq=1 ttl=124 time=69.6 ms
64 bytes from 10.2.0.234: icmp_seq=2 ttl=124 time=68.2 ms
64 bytes from 10.2.0.234: icmp_seq=3 ttl=124 time=68.0 ms
64 bytes from 10.2.0.234: icmp_seq=4 ttl=124 time=68.2 ms
64 bytes from 10.2.0.234: icmp_seq=5 ttl=124 time=68.1 ms
64 bytes from 10.2.0.234: icmp_seq=6 ttl=124 time=68.0 ms
^C64 bytes from 10.2.0.234: icmp_seq=7 ttl=124 time=68.1 ms
64 bytes from 10.2.0.234: icmp_seq=8 ttl=124 time=68.1 ms
64 bytes from 10.2.0.234: icmp_seq=9 ttl=124 time=68.1 ms
64 bytes from 10.2.0.234: icmp_seq=10 ttl=124 time=68.0 ms
64 bytes from 10.2.0.234: icmp_seq=11 ttl=124 time=68.1 ms
64 bytes from 10.2.0.234: icmp_seq=12 ttl=124 time=68.0 ms
^C
--- 10.2.0.234 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11014ms
rtt min/avg/max/mdev = 67.996/68.209/69.643/0.435 ms
[ec2-user@ip-10-0-0-137 ~]$
```

Challenge-2

Connecting from Virginia to Oregon 

```
[ec2-user@ip-10-0-0-137 ~]$ ping 10.3.0.138
PING 10.3.0.138 (10.3.0.138) 56(84) bytes of data.
64 bytes from 10.3.0.138: icmp_seq=1 ttl=124 time=68.3 ms
64 bytes from 10.3.0.138: icmp_seq=2 ttl=124 time=64.9 ms
64 bytes from 10.3.0.138: icmp_seq=3 ttl=124 time=64.9 ms
64 bytes from 10.3.0.138: icmp_seq=4 ttl=124 time=64.8 ms
64 bytes from 10.3.0.138: icmp_seq=5 ttl=124 time=64.9 ms
^C
--- 10.3.0.138 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4003ms
rtt min/avg/max/mdev = 64.825/65.558/68.268/1.355 ms
[ec2-user@ip-10-0-0-137 ~]$
```